

Review Article

Decoding the influence of mitochondrial Ca^{2+} regulation on neurodegenerative disease progression

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ABSTRACT

Mitochondria are pivotal hubs in maintaining cellular homeostasis, encompassing vital processes such as bioenergetics, redox regulation, Ca^{2+} signaling, and programmed cell death. Ca^{2+} is a key second messenger within cells, paramount in numerous critical biological processes. The maintenance of mitochondrial calcium homeostasis relies on a delicate balance between Ca^{2+} uptake and efflux. At the mitochondrial level, Ca^{2+} serves a dual function, participating in essential physiological processes such as ATP production and the regulation of mitochondrial metabolisms and contributing to pathophysiological events, including cell death and cancer metastasis. Alterations in mitochondrial Ca^{2+} ($\text{Ca}_{\text{mito}}^{2+}$) levels influence cellular activity and functionality. The regulation of mitochondrial Ca^{2+} homeostasis involves the collaborative participation of the mitochondrial Ca^{2+} transporter and the mitochondria-endoplasmic reticulum contact sites (MERCs). This review provides a comprehensive overview of current knowledge regarding the regulation of mitochondrial Ca^{2+} homeostasis and its implications in both physiological processes and neurodegenerative disorders. Moreover, we highlight potential opportunities and challenges in developing therapeutic interventions that target mitochondrial Ca^{2+} homeostasis and its regulators, such as novel drug delivery systems and specific calcium-modulating agents.

1. Introduction

Ca^{2+} acts as an important versatile ion signal regulating various intrinsic cellular pathways across different cell types. Ca^{2+} signals are crucial for synaptic transmission, interneuron communication, and neuron-glia interaction within the central nervous system.¹ Neuronal Ca^{2+} influx occurs primarily through localized plasma membrane channels. It is rapidly sequestered by buffers, Ca^{2+} -binding proteins, and internal organelles such as the endoplasmic reticulum (ER) and mitochondria, stimulating ATP production and maintaining the energy requirements for Ca^{2+} homeostasis. Given the inherent capacity of mitochondria to universally uptake, retain, and efflux Ca^{2+} during neuronal activity, this review primarily focuses on the potential links

between mitochondrial Ca^{2+} homeostatic regulators and neurodegenerative disorders.

The mitochondria facilitate the transport of Ca^{2+} from the inner mitochondrial membrane to the matrix via the mitochondrial Ca^{2+} uniporter (MCU) mechanism (Fig. 1). The mild pathology in MCU-deficient animals suggests alternative mitochondrial Ca^{2+} uptake systems or rapid compensatory mechanisms in the absence of MCU.² The regulation of the mitochondrial function is intricately governed by the presence of mitochondrial Ca^{2+} ($\text{Ca}_{\text{mito}}^{2+}$). Within the matrix, $\text{Ca}_{\text{mito}}^{2+}$ exerts precise control over the tricarboxylic acid (TCA) cycle activity and amplifies metabolic output. Specifically, increased levels of Ca^{2+} in the mitochondrial matrix enhance the activity of key enzymes involved in the TCA cycle, such as isocitrate dehydrogenase and α -ketoglutarate

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dehydrogenase. This stimulation leads to augmented substrate availability for oxidative phosphorylation, thereby promoting ATP production. Moreover, Ca^{2+} functions as a pivotal signaling molecule that facilitates the integration of metabolic pathways by linking energy demand with nutrient availability. However, excessive accumulation of $\text{Ca}_{\text{mito}}^{2+}$ disrupts mitochondrial respiration, intensifies reactive oxygen species (ROS) generation, and triggers cell death through the activation of the mitochondrial permeability transition pore (mPTP).³ Mitochondria-endoplasmic reticulum contact sites (MERCs), which serve as dynamic membrane contacts between the endoplasmic reticulum (ER) and the mitochondria, are bridged by highly Ca^{2+} -selective uniporters (MCUs) that facilitate substantial influx of Ca^{2+} into the mitochondria. Mitochondrial Ca^{2+} efflux is achieved by ion exchangers. It is clear that perturbation or disruption of mitochondrial Ca^{2+} homeostasis may lead to pathological conditions.

Neurodegenerative diseases exhibit a myriad of cellular characteristics, encompassing neuronal loss, impaired neuronal function,

aggregation of diverse pathological proteins, mitochondrial dysfunction, oxidative stress, neuroinflammation, and cell death. Beyond ATP production, mitochondria are essential in regulating intracellular Ca^{2+} signaling, cell death, ROS signaling, and maintaining cellular quality control.³ Mitochondrial dysfunction and metabolic disturbances have been implicated in the pathogenesis of various neurodegenerative disorders, including Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD).⁴ Currently, the majority of therapeutic approaches to neurodegenerative disorders primarily focus on the clearance of pathological proteins to alleviate symptoms, while there is a dearth of drug development targeting the inhibition of neurodegeneration.⁵ Imbalances in mitochondrial Ca^{2+} homeostasis or impaired brain energy are thought to occur in the asymptomatic stages of the disease before clinical symptoms,⁵ supporting the hypothesis that impairments in mitochondrial metabolisms may play a pivotal role, if not serve as a determining factor, in the pathogenesis of neurodegenerative disorders. As ongoing research continues to deepen

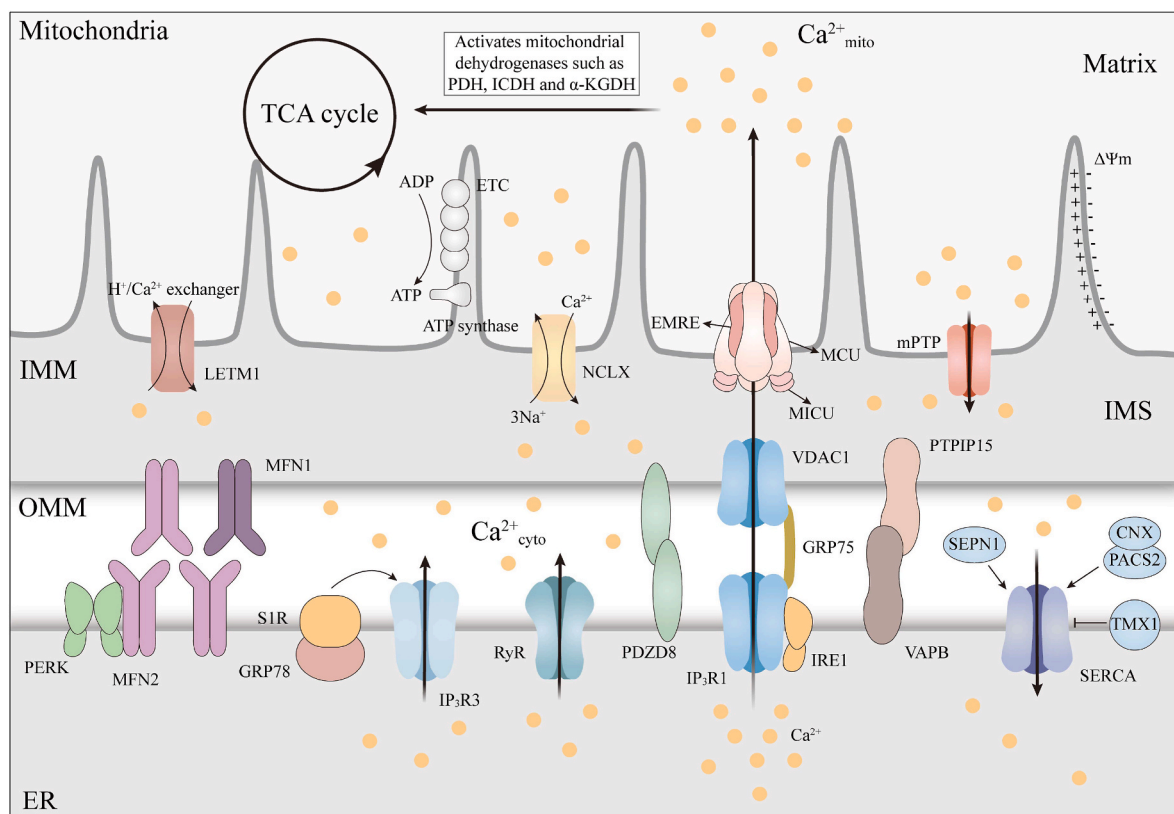


Fig. 1. Regulation of mitochondrial calcium homeostasis and its physiological significance. The outer mitochondrial membrane (OMM) and the inner mitochondrial membrane (IMM) delimitate the intermembrane space (IMS) and the mitochondrial matrix. The communication between the endoplasmic reticulum (ER) and mitochondria is primarily regulated by inositol 1,4,5-trisphosphate receptors 1 to 3 (IP3Rs) and voltage-dependent anion channels (VDACs), which are stabilized and bridged by heat shock protein family A member 9 (GRP75). Ca^{2+} traverses the OMM and enters the IMS through the VDAC channel family. Subsequently, Ca^{2+} flows into the mitochondrial matrix MCU complex. Ca^{2+} efflux from the matrix through the $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger (NCLX). Excessive Ca^{2+} levels lead to mitochondrial Ca^{2+} ($\text{Ca}_{\text{mito}}^{2+}$) overload and the mitochondrial permeability transition pore (mPTP) activation. The transient opening of mPTP in low conductance mode plays a crucial role in maintaining calcium homeostasis. The ER Ca^{2+} load is determined by the sarco-ER Ca^{2+} transport ATPases (SERCAs) and intraluminal Ca^{2+} -binding proteins such as calreticulin, calnexin (CNX), and GRP78/BiP. CNX targets MAM through palmitoylation and phosphofuranic acid cluster sorting protein 2 (PACS2), thereby enhancing SERCA2b activity to facilitate ER Ca^{2+} uptake. Furthermore, selenoprotein N (SEPN1) interacts with SERCA2b and augments its ER Ca^{2+} uptake activity. Mitochondrial fusion factor mitofusin (MFN) maintains mitochondrial network stability by regulating fusion and fission events. PDZ domain containing 8 (PDZD8)-dependent ER-mitochondria contacts play a crucial role in maintaining mitochondrial calcium homeostasis within the dendrites of cortical pyramidal neurons. Vesicle-associated membrane protein-associated protein B (VAPB) and regulatory microtubule dynamics protein 3 (RMDN3, also known as PTPIP51), which amplifies the transfer of Ca^{2+} to mitochondria. The eukaryotic translation initiation factor 2 α kinase 3 (EIF2AK3; also known as PERK) and transducers ER to nucleus signaling 1 (ERN1; also known as IRE1) serve as essential constituents of the unfolded protein response (UPR), playing a pivotal role in upholding the functional integrity of mitochondria. The sigma-1 receptor (S1R) is facilitated by its release from the BiP/GRP78, thereby enhancing ER-to-mitochondria Ca^{2+} transfer by stabilizing IP3R3. As cellular energy demand increases, the elevation in cytosolic Ca^{2+} levels will result in an upregulation of mitochondrial Ca^{2+} concentration, triggering the activation of three vital mitochondrial dehydrogenases: pyruvate dehydrogenase (PDH), isocitrate dehydrogenase (ICDH), and α -ketoglutarate dehydrogenase (α -KGDH). Activating these dehydrogenases by Ca^{2+} leads to an augmented NADH production, which transfers electrons to the electron transport chain (ETC).

comprehension of mitochondrial Ca^{2+} dysregulation as an early diagnostic tool or pathogenic mechanism for neurodegenerative disorders, the targeted modulation of specific mitochondrial Ca^{2+} transportation channels and regulation of ER-mitochondria contacts may hold clinical significance in ameliorating neurodegeneration.

In this review, we initially present an overview of the regulation of mitochondrial Ca^{2+} homeostasis and its physiological significance. Subsequently, we elucidate the alterations and distinctive features of Ca^{2+} signaling in various neurodegenerative diseases, thereby offering potential applications for targeting different regulatory mechanisms of mitochondrial Ca^{2+} as means for early diagnosis and treatment of neurodegenerative disorders.

2. Regulation of mitochondrial Ca^{2+} homeostasis and its physiological significance

Mitochondria are complex, bilayered organelles that are present in nearly all eukaryotic cell types. The outer mitochondrial membrane (OMM) and inner mitochondrial membrane (IMM) define two distinct compartments, the intermembrane space (IMS) located between these membranes and the mitochondrial matrix, which lies within the inner mitochondrial membrane.⁶ Neurons rely on mitochondrial oxidative respiration for ATP production and synaptic transmission. The preservation of mitochondrial functional integrity is imperative for the sustenance of processes such as synaptic plasticity, cognition, and the retention of memories.⁷ Mitochondria dynamically regulate cellular Ca^{2+} signaling by buffering cytosolic Ca^{2+} . The MCU complex selectively transports Ca^{2+} into the mitochondrial matrix, while Ca^{2+} efflux is primarily achieved by the mitochondrial sodium- Ca^{2+} exchanger NCLX.^{8,9} Furthermore, the MERCS represent specialized regions where the ER establishes close physical proximity with the OMM^{10,11} (Fig. 1). At these contact sites, ER Ca^{2+} channels facilitate the transfer of Ca^{2+} into the mitochondria. Mitochondrial Ca^{2+} homeostasis is maintained by a dynamic equilibrium between Ca^{2+} influx and efflux. The balance between Ca^{2+} entry through the MCU and the efflux via the NCLX ensures optimal mitochondrial function and prevents Ca^{2+} overload. This section summarizes the regulatory factors governing mitochondrial Ca^{2+} homeostasis and elucidates the physiological significance of mitochondrial Ca^{2+} signaling.

2.1. Mitochondrial Ca^{2+} uptake

ER-mitochondria communication is primarily regulated by inositol 1,4,5-trisphosphate receptors type 1 to 3 (IP₃Rs) and voltage-dependent anion channels (VDACs), stabilized and bridged by heat shock protein family A member 9 (HSPA9; also known as GRP75).¹² VDACs are functionally coupled to the MCU complex in the IMM. Ca^{2+} traverses the OMM and enters the IMS through the VDAC channel family.¹³ Among several Ca^{2+} -sensitive proteins involved in cytosolic and mitochondrial redox homeostasis and metabolism, the Ca^{2+} -binding mitochondrial carrier protein citrin is located in the IMS and functions as part of the malate-aspartate shuttle to regulate essential import functions in mitochondrial biology.¹⁴ Subsequently, Ca^{2+} flows into the mitochondrial matrix via uniporters in the IMM.¹⁵ The MCU complex consists of four fundamental constituents: the pore-forming MCU protein, an indispensable auxiliary subunit essential MCU regulator (EMRE), and the vigilant gatekeepers mitochondrial Ca^{2+} uptake (MICU) 1 and MICU2/3.^{16,17} EMRE functions as a Ca^{2+} sensor on both sides of the IMM, preventing excessive Ca^{2+} accumulation and depletion.¹⁸ MICU1 and MICU2 serve as regulatory gatekeepers of the MCU, preventing the uptake of Ca^{2+} by mitochondria under resting conditions.¹⁹

Transferring Ca^{2+} from the IMS to the mitochondrial matrix through the MCU complex is an exquisitely intricate process. The opening of the MCU complex exhibits a sigmoidal dependence on Ca^{2+} uptake, characterized by a low affinity for Ca^{2+} .²⁰ The uptake of Ca^{2+} by the MCU complex is dynamically regulated. Upon entry into the mitochondrial

matrix, Ca^{2+} reduces the transmembrane potential difference, attenuating the driving force for Ca^{2+} influx.²¹ Although the MCU serves as the principal mechanism for mitochondrial Ca^{2+} influx, it has been reported that other transport systems, such as mitochondrial ryanodine receptors in cardiomyocytes and neurons, also contribute to the uptake of mitochondrial Ca^{2+} .²² Ca^{2+} entry into the mitochondria enhances oxidative phosphorylation by stimulating the activity of rate-limiting dehydrogenases within the TCA cycle.²³ Constitutive IP₃R-mediated Ca^{2+} release and subsequent VDAC translocation into mitochondria are essential for the maintenance of fundamental bioenergetics under resting conditions.¹²

2.2. Mitochondrial matrix Ca^{2+} efflux

Ca^{2+} efflux from the mitochondrial matrix is mediated by specific exchangers in the IMM (Fig. 1). Two primary types of Ca^{2+} exchangers have been identified, $\text{H}^{+}/\text{Ca}^{2+}$ exchangers (mH₁CX) and $\text{Na}^{+}/\text{Ca}^{2+}$ exchangers (mNCX). $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger activity is executed by the protein NCLX, a transporter that exchanges three Na^{+} for one Ca^{2+} .²⁴ Remarkably, Li^{+} can substitute for Na^{+} in the $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger to facilitate the efflux of Ca^{2+} from the mitochondrial matrix.²⁵ Functional analysis of mitochondrial Ca^{2+} transport reveals that NCLX overexpression enhances Ca^{2+} efflux, while silencing NCLX impairs mitochondrial Ca^{2+} transfer.²⁶ The activity of the $\text{H}^{+}/\text{Ca}^{2+}$ exchanger is attributed to the leucine zipper EF-hand-containing transmembrane (LETM1) protein.²¹ Although direct evidence is currently lacking, LETM1 may modulate NCLX activity by influencing the osmotic balance rather than directly regulating the electrical exchange of H^{+} and Ca^{2+} .²⁷ Ca^{2+} signals undergo continuous conversion within the ER and mitochondria through the action of uniporters and exchangers, producing observable Ca^{2+} oscillations in these organelles.²¹

Under physiological conditions, mitochondrial Ca^{2+} homeostasis can be restored through the mitochondrial exchanger, while the excessive influx of Ca^{2+} or impaired efflux of Ca^{2+} may trigger mitochondrial Ca^{2+} overload.²⁸ Excessive Ca^{2+} levels increase IMM permeability through the activation of the mitochondrial permeability transition pore (mPTP).²⁹ Continuous mPTP opening causes ion, protein, and apoptotic molecule efflux, leading to mitochondrial swelling, impaired ATP production, and cell death.³⁰ An mPTP operates in low and high conductance modes, mediating its physiological effects. The high conductance mode occurs in pathological conditions, allowing macromolecule permeability, dissipating transmembrane potential, and triggering apoptosis.³¹ Transient mPTP opening in low conductance mode helps maintain Ca^{2+} homeostasis by releasing Ca^{2+} from the mitochondrial matrix.²¹ Therefore, the low conductance mode will effectively avoid mitochondrial Ca^{2+} overload without damaging the bioenergetics of releasing mitochondrial metabolic substrates.³²

2.3. Regulatory factors of mitochondrial matrix Ca^{2+} homeostasis

This section provides an in-depth analysis of regulatory factors and structural determinants influencing mitochondrial Ca^{2+} homeostasis. The MCU complex is subject to regulation by several accessory proteins, including MICU1/2 and EMRE.^{16,17} The MCU paralog MCUB function as a dominant negative protein in the complex.³³ MCU regulator 1 (MCUR1) serves as a scaffold, connecting MCU and EMRE to regulate the Ca^{2+} permeability of MCU.³⁴ Mitochondria derived from MCU-KO mice brains exhibit diminished efficiency in Ca^{2+} uptake.³⁵ In WT mice mitochondria, over elevated Ca^{2+} levels induced swelling and mPTP opening, an effect that was absent in mitochondria from MCU knockout (MCU-KO) mice.³⁵ Hence, further investigation is warranted to explore the regulatory mechanisms and pathophysiological implications of the MCU complex in maintaining Ca^{2+} homeostasis across diverse disease models.

The mitochondrial $\text{Na}^{+}/\text{Ca}^{2+}$ exchanger crosstalks with cell membrane channels through multiple pathways. Store-operated Ca^{2+} entry

(SOCE) increases cytosolic Na^+ , enhancing Na^+ uptake and Ca^{2+} efflux by NCLX.³⁶ Ca^{2+} indirectly modulates the $\text{Na}^+/\text{Ca}^{2+}$ exchanger by influencing mitochondrial membrane potential ($\Delta\Psi_m$) and MCU activity.³⁷ However, it is unclear whether Ca^{2+} regulates exchanger activity directly or indirectly in an allosteric manner. Furthermore, the activity of the mitochondrial $\text{Na}^+/\text{Ca}^{2+}$ exchanger is also modulated by various regulatory proteins and kinases. Tetanic stimulation of motor neurons increases neurotransmitter release via Na^+ -dependent mitochondrial Ca^{2+} release, inhibited by protein kinase C (PKC) inhibitors, suggesting PKC enhances NCLX activity.³⁸

In a PD model with PTEN-induced putative kinase 1 (PINK1) mutations, Ca^{2+} overload in PINK1-deficient cells is linked to impaired Ca^{2+} efflux mediated by the loss of mitochondrial NCLX function.³⁹ Further investigations have revealed that protein kinase A (PKA) effectively restores NCLX activity by phosphorylating serine 258 (Ser258).⁴⁰ Notably, the NCLX S258D mutant prevents mitochondrial Ca^{2+} overload and mitochondrial depolarization in PINK1-KO neurons, thereby improving neuronal survival.⁴⁰ Additionally, leucine-rich repeat kinase 2 (LRRK2) mutations are associated with the late onset of PD. Expression of mutant LRRK2 resulted in the transcriptional upregulation of mitochondrial MCU and MICU1, leading to excessive accumulation of mitochondrial Ca^{2+} , while the expression levels of NCLX remained unaltered.⁴¹ Interestingly, the expression of the NCLX S258D mutant significantly enhanced Ca^{2+} efflux and effectively rescued neurons from mitochondrial Ca^{2+} overload.⁴¹ Collectively, modulation of NCLX activity at Ser258 through mechanisms such as phosphorylation or phosphomimetic mutations may serve as a potential therapeutic strategy to mitigate the detrimental cycle of mitochondrial Ca^{2+} overload observed in neurodegenerative disorders.^{39–42}

Ca^{2+} flux from the ER to the mitochondria is essential for maintaining mitochondrial Ca^{2+} homeostasis. ER intraluminal Ca^{2+} concentration is important for initiating precise Ca^{2+} signaling and ensuring proper protein folding and maturation. STIM detects ER Ca^{2+} depletion and directly binds to Orai, thereby facilitating the opening of Ca^{2+} -release-activated calcium (CRAC) channels and initiating influx of Ca^{2+} .⁴³ Additionally, transmembrane and coiled-coil domains 1 (TMCO1) senses excessive ER Ca^{2+} levels and forms homotetrameric Ca^{2+} channels, leading to the release of Ca^{2+} from ER store.⁴⁴ Ca^{2+} homeostasis is also maintained by tight ER-mitochondria connections and the ER Ca^{2+} load (Fig. 1). The ER Ca^{2+} load is determined by sarco-ER Ca^{2+} transport ATPases (SERCAs) and intraluminal Ca^{2+} -binding proteins such as calreticulin, calnexin (CNX), and GRP78/BiP, which effectively maintain elevated levels of Ca^{2+} within the ER.⁴⁵ CNX promotes ER Ca^{2+} uptake by targeting SERCA2b through palmitoylation and the phosphofuranic acid cluster sorting protein 2 (PACS2).⁴⁶ Furthermore, selenoprotein N (SEPN1) has been demonstrated to interact with SERCA2b, augmenting its ER Ca^{2+} uptake activity and protecting cells from ROS production during oxidative protein folding.⁴⁷ Significantly, deletion of the thioredoxin-related transmembrane protein 1 (TMX1) enhances ER Ca^{2+} uptake rates, augmenting the probability of Ca^{2+} transfer to the mitochondria.^{48,49}

The role of the ER in Ca^{2+} signaling is facilitated by its remarkable capacity for calcium storage: the concentration of free and protein-bound Ca^{2+} within the ER lumen exceeds that in the cytoplasm by three orders of magnitude. As a Ca^{2+} reservoir, the ER regulates mitochondrial Ca^{2+} homeostasis in various modes. Besides its involvement in ER Ca^{2+} uptake, ER-mitochondria contact modulates Ca^{2+} transport. The IP₃R1-GRP75-VDAC1 complex regulates Ca^{2+} transfer from the ER to the mitochondria. IP₃R modulates Ca^{2+} signaling, influencing processes such as embryonic development and neuronal plasticity.⁵⁰ GRP75 knockdown attenuates Ca^{2+} transfer from the ER to the mitochondria, whereas GRP75 upregulation promotes MERCs formation, resulting in excessive mitochondrial Ca^{2+} accumulation and subsequent impairment of neuronal mitochondrial function.⁵¹ The mitochondrial fusion factor mitofusin (MFN) is a mitochondrial GTPase involved in Ca^{2+} regulation, including MFN1 and MFN2. MFN2 maintains mitochondrial network

stability by regulating fusion and fission events. Furthermore, MFN2 on the ER interacts with MFN1/2 on the OMM and facilitates the formation of a connexin complex, forming a connexin complex for efficient ER-to-mitochondria Ca^{2+} transfer. MFN2 ablation results in reduced ER-mitochondria contacts.^{52,53} Additionally, the PDZ domain containing 8 (PDZD8)-dependent ER-mitochondria contacts are crucial for mitochondrial Ca^{2+} homeostasis in cortical pyramidal neuron dendrites. Conversely, PDZD8 loss causes ER Ca^{2+} release into the cytoplasm instead of the mitochondria.⁵⁴ Other contributing factors of significance in establishing the connection between the ER and mitochondria include a complex formed by the vesicle-associated membrane protein-associated protein B (VAPB) and regulatory microtubule dynamics protein 3 (RMDN3, also known as PTPIP51), which amplifies the transfer of Ca^{2+} and lipids to the mitochondria.⁵⁵

The cellular process of the UPR is utilized to optimize proteostasis and regulate lipid synthesis, thereby restoring Ca^{2+} concentrations and maintaining cellular redox homeostasis.⁵⁶ The involvement of UPR in modulating mitochondrial Ca^{2+} homeostasis at ER-mitochondria contacts is discussed below (Fig. 1). The eukaryotic translation initiation factor 2 α kinase 3 (EIF2AK3, also known as PERK) and transducers ER to nucleus signaling 1 (ERN1, also known as IRE1) serve as essential constituents of the UPR, playing a pivotal role in upholding the functional integrity of the mitochondria.^{57,58} Studies have shown that PERK-deficient cells exhibit disordered ER morphology and Ca^{2+} signaling and significantly weaker ER-mitochondria contact sites.⁵⁷ The interaction between MFN2 and PERK facilitates the inhibitory effect of MFN2 on PERK activity during both resting and ER stress conditions, thereby mitigating ROS production, restoring mitochondrial Ca^{2+} homeostasis, and enhancing mitochondrial morphology.⁵⁹ The activity of PERK and the α subunit of eukaryotic initiation factor 2 (eIF2 α)/activating transcription factor 4 (ATF4) enhances the expression of mitochondrial and MERCs components, including the truncated isoform of ATPase sarcoplasmic reticulum/ER Ca^{2+} transporter 1 (S1T), parkin RBR E3 ubiquitin-protein ligase (PRKN; also known as PARKIN), or sigma-1 receptor (S1R).^{60,61} IRE1 is a scaffold to dock IP₃R on MERCs to promote Ca^{2+} transfer to the mitochondria.⁵⁸ IRE1 activity in MERCs is stabilized by the presence of S1R and attenuated by the action of the mitochondrial E3 ubiquitin ligase membrane-associated ring-CH-type finger 5 (MARCHF5, also known as MITOL).⁶² Moreover, S1R enrichment at MERCs is facilitated by its release from the Ca^{2+} -dependent molecular chaperone BiP/GRP78, enhancing ER-to-mitochondria Ca^{2+} transfer by stabilizing IP₃R3 during low ER Ca^{2+} levels.⁶³ MITOL specifically interacts with and ubiquitinates mitochondrial MFN2 at residue K192, thereby facilitating the activation of MFN2 to regulate the tethering between the ER and the mitochondria.⁶⁴ It can be inferred that a close interdependence exists between the maintenance of ER homeostasis and MERCs, a function that may also be relevant in neurodegenerative disorders.⁶⁵ Notably, IRE1 has also been reported to mitigate mitochondrial Ca^{2+} overload by interacting with IP₃R in neuronal cell lines.⁶⁶ Additionally, studies have identified mitochondrial proteins that bind to IRE1, such as ATP5H, to promote IRE1 signal transduction.⁶⁷

In summary, perturbations in mitochondrial Ca^{2+} dynamics may be associated with various pathological conditions, including disorders in mitochondrial Ca^{2+} uptake and efflux and alterations in mitochondrial molecular components, and disruptions in ER-mitochondria contacts can lead to impairment of cell bioenergetics and trigger different neurodegenerative diseases.^{21,68}

2.4. The pathophysiological role of mitochondrial Ca^{2+} signaling

Mitochondrial Ca^{2+} is essential for ATP synthesis and cellular energy production. Mitochondrial Ca^{2+} uptake and efflux regulate matrix Ca^{2+} homeostasis and modulate mitochondrial dehydrogenase activity.¹² When cellular energy demand increases, elevated cytosolic Ca^{2+} levels increase mitochondrial Ca^{2+} concentration, activating three key dehydrogenases: pyruvate dehydrogenase (PDH), isocitrate

dehydrogenase (ICDH), and α -ketoglutarate dehydrogenase (α -KGDH).³³ Activating these dehydrogenases by Ca^{2+} leads to an augmented NADH production, which transfers electrons to the electron transport chain (ETC) (Fig. 1). The IMM contains the electron transport complexes I–IV of the oxidative phosphorylation system (OXPHOS).⁶⁹ Complexes (I–IV) utilize the energy from electrons to translocate protons from the matrix to the IMS, while complex V functions as an F_1F_0 -ATP synthase that harnesses energy from the electrochemical gradient for ATP synthesis.⁷⁰ The mitochondrial respiratory chain triggers the electrochemical gradient waves required to produce ATP through five enzyme complexes, besides generating the by-product ROS. Elevated Ca^{2+} levels increase ROS production in the respiratory chain. Excessive ROS and Ca^{2+} can trigger mPTP opening, releasing pro-apoptotic factors and initiating apoptosis.⁷¹

Excessive ROS can directly impair Ca^{2+} regulatory proteins, resulting in elevated Ca^{2+} concentrations and disruption of Ca^{2+} homeostasis.⁷² H_2O_2 generated from Ca^{2+} overload at ER-mitochondria contacts modulates agonist-induced Ca^{2+} release and oscillations.⁷³ Oxidative stress and Ca^{2+} homeostasis imbalance induce ER stress, activating the UPR to mitigate the aberrant accumulation of misfolded proteins within cells and restore ATP production. During ER stress, the PERK-eIF2 α -ATF4 axis enhances the assembly factor 1 (SCAF1), facilitating the formation of respiratory supercomplexes (SC) and augmenting mitochondrial respiration.⁷⁴ IRE1 α is a key regulator of a specific branch within the UPR pathway, activated in response to ER stress, thereby facilitating the restoration of ER proteostasis.⁷⁵ Furthermore, comprehensive proteomic and metabolomic investigations have revealed that a deficiency of IRE1 α in hepatic tissue disrupts crucial metabolite levels associated with the TCA cycle.⁷⁶ Thus, core UPR components such as PERK and IRE1 regulate ER-mitochondria contacts and play an indispensable role in maintaining mitochondrial biological functions to regulate ATP production.

Synaptic Ca^{2+} precisely regulates mitochondrial function and dynamics at synapses. Mitochondrial Ca^{2+} buffering during synaptic activity protects against synaptic transmission impairment and neuronal excitotoxicity. Mitochondrial Ca^{2+} uptake modulates neuronal transport and acts as a buffer, sequestering and releasing Ca^{2+} to maintain synaptic integrity.⁷⁷ Mitochondrial Ca^{2+} uptake in synaptic transmission actively promotes ATP synthesis, which is essential for the docking, fusion, and release of neurotransmitter-containing vesicles during synaptic transmission.⁷⁸ Mitochondrial Ca^{2+} influx is critical for increasing ATP production to meet the energy demands associated with synaptic transmission. Furthermore, regulating Ca^{2+} release in the ER governs many neuronal functions, ranging from modulation of plasma membrane excitability to synaptic plasticity.⁷⁹ Additionally, the mitochondria connect with the ER, facilitating signal transduction and biosynthetic pathways, influencing organelle distribution, and mediating Ca^{2+} signaling propagation towards the mitochondria.⁷⁹ Excessive glutamate release induces neuronal excitotoxicity, elevating intracellular Ca^{2+} and overwhelming mitochondrial Ca^{2+} buffering.⁷⁷ Thus, while mitochondrial Ca^{2+} uptake is critical for the activity-dependent promotion of ATP generation during synaptic transmission, sustained neuronal Ca^{2+} elevation impairs ER-mitochondrial Ca^{2+} regulation, potentially leading to neuronal death.

3. Ca^{2+} homeostasis regulation in neurodegenerative diseases

Neurodegenerative diseases are characterized by neuronal loss, protein aggregation, disrupted mitochondrial Ca^{2+} homeostasis, oxidative stress, neuroinflammation, and cell death. Beyond ATP production, mitochondria regulate intracellular Ca^{2+} signaling, cell death, ROS signaling, and cellular quality control.⁸⁰ Mitochondrial dysfunction and metabolic disruption have been implicated in neurodegenerative diseases such as Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD), amyotrophic lateral sclerosis (ALS), and spinocerebellar ataxia (SCA).^{4,81} The next section will discuss the

cellular functions of mitochondrial Ca^{2+} and delineate the pivotal role of mitochondrial Ca^{2+} homeostasis and its regulatory factors in neurodegenerative disorders, highlighting the molecular underpinnings of mitochondrial function and metabolism in diverse neurodegenerative diseases.

3.1. Alzheimer's disease

Alzheimer's disease is an age-related neurodegenerative disorder characterized by progressive memory impairment, cognitive dysfunction, language deficits, and other associated symptoms. The pathogenesis of AD involves extensive amyloid beta ($\text{A}\beta$) deposition, forming senile plaques in the brain parenchyma, and the intracellular accumulation of hyperphosphorylated tau protein, forming neurofibrillary tangles (NFTs).⁸² $\text{A}\beta$ aggregation initiates a cascade of events, including tau pathology, oxidative stress, inflammation, disrupted neuronal Ca^{2+} homeostasis, and metabolic disturbances, ultimately leading to neuronal cell loss and reduced synaptic density.⁸³ In addition to $\text{A}\beta$ deposition, synaptic degeneration, NFT formation, mitochondrial dysfunction, and disrupted Ca^{2+} signaling are crucial events in AD.

Alzheimer's disease patients often show sporadic deficiencies in ETC complexes I and IV, reduced $\Delta\psi\text{m}$, altered mitochondrial morphology, increased $\text{A}\beta$ deposition, and tau oligomerization⁸⁴ (Fig. 2). Furthermore, compared to healthy controls, fibroblasts derived from late-onset AD patients exhibit a metabolic shift from mitochondrial OXPHOS towards glycolysis, indicating a diminished mitochondrial metabolic capacity in individuals with late-onset AD.⁸⁵ Both $\text{A}\beta$ and tau proteins can be transported to the mitochondrial matrix through translocase of the outer mitochondrial membrane (TOM) 40 and translocase of the inner mitochondrial membrane (TIM) 23, where they interact with ETC, thereby affecting the normal function of the mitochondria and ATP production.⁸⁶ Postmortem brain tissue obtained from AD patients exhibit decreased PDH, α -KGDH, and ICDH enzyme activities.⁸⁷ Impaired PDH function restricts pyruvate entry into the TCA cycle, causing pyruvate accumulation and promoting anaerobic metabolism.⁸⁸ Lactic acid accumulation from glycolysis reduces acetyl-CoA availability, impairing OXPHOS. Defective mitochondrial pyruvate flux, associated with overactive glycogen synthase kinase-3 β (GSK3 β), impairs hexokinase 1 recruitment to the mitochondria and destabilizes the mitochondrial pyruvate carrier complex.⁸⁹ Interestingly, the activities of these key enzymes are intricately regulated by Ca^{2+} , underscoring the crucial role of neuronal mitochondrial Ca^{2+} homeostasis in AD pathogenesis, which will be discussed further.

Neuronal communication requires Ca^{2+} influx through voltage-dependent Ca^{2+} channels for synaptic vesicle fusion and neurotransmitter release. This energy-intensive process relies on mitochondrial ATP generation via OXPHOS and maintenance of synaptic Ca^{2+} homeostasis. Any dysfunction in mitochondrial Ca^{2+} uptake and efflux may impair mitochondrial Ca^{2+} homeostasis maintenance, leading to mitochondrial Ca^{2+} overload, oxidative stress, and subsequent neuronal death.⁹⁰ The formation of $\text{A}\beta$ oligomers in the extracellular space elicits excessive activation of Ca^{2+} channels, leading to a substantial influx of Ca^{2+} into neurons, thereby perturbing Ca^{2+} homeostasis and provoking extensive ROS generation.⁹¹ Elevated intracellular Ca^{2+} activates calcineurin, which, along with $\text{A}\beta$ or tau, induces dendritic spine and synaptic loss, contributing to cognitive impairments in AD mouse models.⁹²

The ER and the mitochondria play essential roles in regulating intracellular Ca^{2+} signaling in healthy individuals. Pathologically, $\text{A}\beta$ induces the ER to release Ca^{2+} into the cytoplasm by activating IP $_3$ R and ryanodine receptors (RyRs) (Fig. 2). SERCAs are integral proteins in the ER, which are responsible for actively transporting free Ca^{2+} into the ER lumen through ATP hydrolysis. Immunoprecipitation analyses have revealed a physical interaction between SERCA and presenilin (PS/PSEN) 1 and PS2, highlighting the significance of SERCA expression modulation in regulating $\text{A}\beta$ levels.⁹³ Overexpressing wild-type or

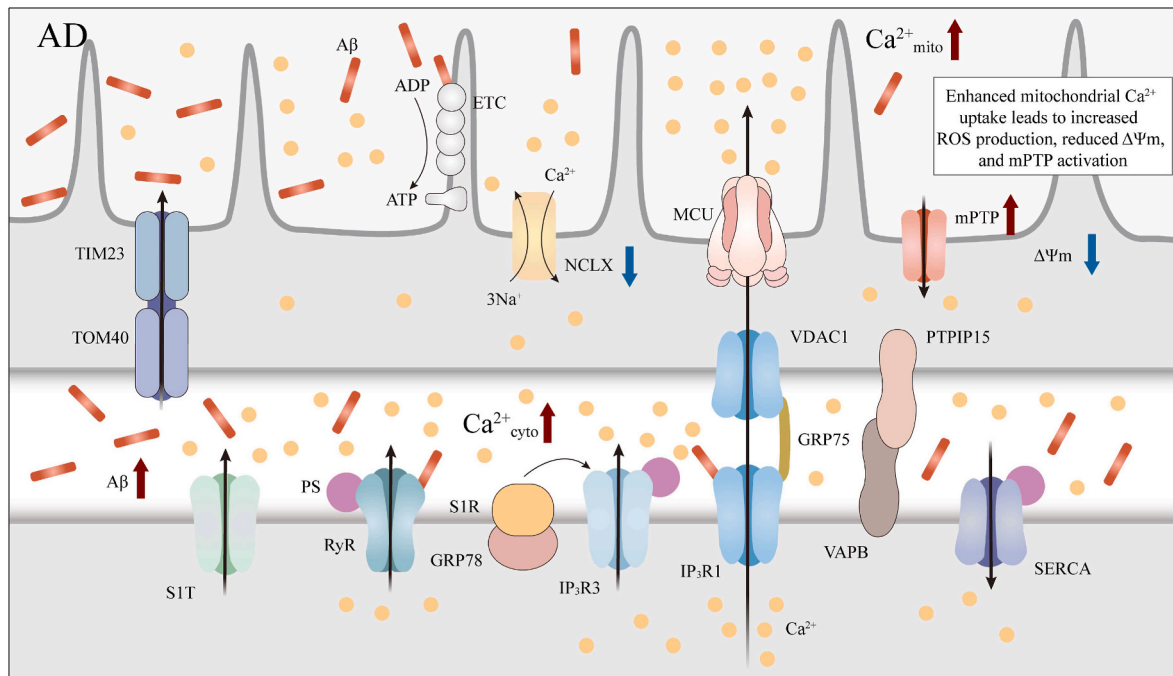


Fig. 2. Calcium homeostasis regulation in AD. Both amyloid beta ($A\beta$) and tau proteins can be transported to the mitochondrial matrix through translocase of the outer mitochondrial membrane (TOM) 40 and translocase of the inner mitochondrial membrane (TIM) 23, where they interact with ETC, thereby affecting the normal function of mitochondria and ATP production. Under pathological conditions, $A\beta$ induces ER to release Ca^{2+} into the cytoplasm (Ca^{2+}_{cyto}) by activating IP3R and ryanodine receptors (RyRs). The overexpression of wild-type or mutant PS2 induces passive ER leakage through IP3R and RyR, which the overexpression of SERCA2b can counteract. The activation of S1T occurs in response to pharmacological and pathophysiological ER stress. $A\beta$ drives mitochondrial Ca^{2+} uptake through the MCU complex, triggering mitochondrial Ca^{2+} overload and leading to increased ROS production, reduced $\Delta\Psi_m$, and mPTP activation, ultimately leading to neuronal death. Another proposed mechanism of $A\beta$ -mediated mitochondrial Ca^{2+} overload is the impairment of Ca^{2+} efflux through reduced NCLX levels.

mutant PS2 induces passive ER Ca^{2+} leakage through IP3R and RyR, counteracted by SERCA2b overexpression.^{94,95} By contrast, inhibiting SERCA and releasing Ca^{2+} into the cytoplasm through IP3R and RyR channels triggers a state of Ca^{2+} overload, activating β -secretase and ultimately leading to an increase in $A\beta$ production and aggregation⁹⁶ (Fig. 2). Overall, SERCA potentially contributes to maintaining ER Ca^{2+} homeostasis in AD models.

Moreover, $A\beta$ triggers ER stress by promoting the accumulation of misfolded proteins within the ER and perturbing Ca^{2+} homeostasis, thereby eliciting UPR⁹⁷ (Fig. 2). S1T activation occurs in response to pharmacological and pathophysiological ER stress, which is mediated by the PERK-eIF2 α -ATF4 pathway.⁹⁸ S1T upregulation has been demonstrated in SH-SY5Y cells expressing amyloid beta precursor protein (APP) swe, and is induced by the exogenous application of $A\beta$ oligomers.⁹⁹ Interestingly, S1T overexpression enhances beta-site APP cleaving enzyme 1 (BACE1) activity, promoting APP processing and $A\beta$ production in 3xTgAD mice.⁹⁹ Elucidating the molecular interplay among ER Ca^{2+} homeostasis, ER stress, and APP processing is important for comprehending AD progression.

Direct exposure of neurons to soluble $A\beta$ oligomers enhances mitochondrial Ca^{2+} uptake, causing Ca^{2+} overload, increased ROS production, reduced $\Delta\Psi_m$, and mPTP activation, leading to neuronal death⁹⁰ (Fig. 2). Specifically, in an APP/PSEN1 mouse model, neurons show elevated mitochondrial Ca^{2+} levels in both soma and neurites compared to wild-type mice.⁹⁰ Mechanistically, $A\beta$ drives mitochondrial Ca^{2+} uptake through the MCU complex (Fig. 2). Elevated mitochondrial Ca^{2+} levels in neurons lead to apoptosis in the APP/PSEN1 mouse model, suggesting an association between these processes, though causality remains undetermined. Furthermore, $A\beta$ enhances Ca^{2+} transfer from the ER to the mitochondria through the MERCS, potentially by reducing the optimal distance between the ER and the mitochondria or increasing their contacts, raising mitochondrial Ca^{2+} levels.¹⁰⁰ In support of this, exposure of hippocampal neurons to $A\beta$ promotes ER-to-mitochondria

Ca^{2+} transfer and disrupts mitochondrial Ca^{2+} homeostasis by increasing ER-mitochondria contact.¹⁰¹ Another proposed mechanism of $A\beta$ -mediated mitochondrial Ca^{2+} overload is the impairment of Ca^{2+} efflux through reduced NCLX levels. Studies have reported a reduction in NCLX expression in the mitochondria from human AD brains, which is associated with compromised mitochondrial Ca^{2+} efflux and subsequent Ca^{2+} overload, potentially contributing to AD progression.¹⁰² On the other hand, hippocampal neurons from mice expressing the apolipoprotein E (APOE) $\epsilon 4$ allele show upregulation of MERCS activity and ER-mitochondria communication, implicating MERCS regulation in AD pathogenesis.¹⁰³ Loss of PS or mutation of PS FAD enhances ER-to-mitochondria Ca^{2+} uptake, indicating that PS modulates ER-mitochondria Ca^{2+} transfer through a mechanism independent of $A\beta$.¹⁰⁴ Taken together, the impact of alterations in MERCS distribution and function on mitochondrial Ca^{2+} homeostasis is likely to be intricately linked to the pathogenesis of AD. The augmented interplay between the ER and the mitochondria observed across various AD models holds significant implications for mitochondrial functionality.

In conclusion, mitochondrial Ca^{2+} homeostasis imbalance is crucial in neuronal cell death, and targeting mitochondrial Ca^{2+} through pharmaceutical interventions may offer a promising treatment for AD. Future research should elucidate the causal connection between mitochondrial Ca^{2+} overload and $A\beta$ plaques formation. Additionally, it is essential to investigate the regulation of mitochondrial Ca^{2+} homeostasis across different cell types in AD brains and various AD models.

3.2. Parkinson's disease

Parkinson's disease is the second most prevalent neurodegenerative disorder, characterized by bradykinesia, rigidity, postural instability, and resting tremor. The main motor symptoms of PD can be attributed to the progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta (SNpc) of the brain, the accumulation of

intracellular inclusions containing α -Synuclein (α -Syn), and subsequent progressive damage to dopaminergic neurons.¹⁰⁵ Mutated forms of α -Syn, particularly those harboring A53T and A30P mutations, reduce MERCS distance, disrupt VAPB/PTPIP51 interactions, and impair mitochondrial Ca^{2+} uptake, potentially attributed to an augmented interaction between mutated α -Syn and VAPB¹⁰⁶ (Fig. 3). However, variations in the relative proportions of wild-type and mutant α -Syn may yield disparate outcomes in MERCS Ca^{2+} transport.¹⁰⁷ α -Syn translocates to the IMM via VDAC, inhibiting Ca^{2+} transport into the mitochondria.¹⁰⁸ Furthermore, studies have demonstrated that α -Syn can disrupt the interaction between IP₃R and GRP75, thereby impeding the transfer of Ca^{2+} from the ER to the mitochondria¹⁰⁹ (Fig. 3). Accumulated pathological α -Syn preferentially binds to mitochondria, hindering mitochondrial protein import and Ca^{2+} uptake, resulting in mitochondrial membrane depolarization and compromised cellular respiration.¹¹⁰ Furthermore, studies have shown that DJ-1 (encoded by PARK7) is also located in the MERCS and can interact with the IP3R-GRP75-VDAC protein complex to promote Ca^{2+} transfer; however, this localization and function are lost in cells expressing PD-associated mutant PARKIN.¹¹¹ A deficiency of DJ-1 disrupts the interaction of the protein complex, resulting in reduced ER-mitochondria contacts¹¹² (Fig. 3). Therefore, reduced DJ-1 levels in the substantia nigra of PD patients can disrupt ER-mitochondria contacts, ultimately leading to mitochondrial structural disorder and dysfunction. Furthermore, studies have demonstrated a direct interaction between DJ-1 and both α -Syn monomers and oligomers.¹¹³ Notably, overexpression of DJ-1 has been shown to effectively mitigate α -Syn dimerization, while inhibition of DJ-1 increases aggregated α -Syn levels in PD patients.¹¹³

PARKIN (encoded by PARK2) and PINK1 (encoded by PARK6) are two genes located in the MERCS that undergo mutations in familial PD. Physiologically, either PINK1 or PARKIN is indispensable for maintaining optimal ER-mitochondria contact. Overexpression of PARKIN in substantia nigra neurons increases ER-mitochondria contacts.¹¹⁴ Furthermore, PINK1 gene silencing in M17 dopaminergic cells reduces ER-mitochondria contacts, impairing mitochondrial Ca^{2+} homeostasis.¹¹⁵ PINK1 and PARKIN deletions or mutations induce ER stress due to impaired mitochondrial-ER association.¹¹⁶ Disruption of communication between dysfunctional mitochondria and the ER can ameliorate signaling via the PERK branch of ER stress¹¹⁶ (Fig. 3). Furthermore, PERK facilitates the ubiquitination and subsequent degradation of the mitochondrial fission factor MFN2 via PINK1 and PARKIN, thereby facilitating efficient clearance of damaged mitochondria⁶⁵ (Fig. 3). Additionally, leucine-rich repeat kinase 2 (LRRK2) mutations are associated with the late onset of PD. Although there is a lack of evidence demonstrating the ability of LRRK2 to induce mitochondrial Ca^{2+} efflux through NCLX, the deficiency-induced inhibition of mitochondrial Ca^{2+} efflux by LRRK2 can be rescued through the upregulation of NCLX¹¹⁷ (Fig. 3).

In conclusion, regulating mitochondrial Ca^{2+} homeostasis and MERCS integrity is pivotal in PD pathogenesis. However, the precise mechanisms governing ER-mitochondria signaling in PD-related disruption of Ca^{2+} homeostasis remain elusive. Therefore, further investigations are warranted to comprehensively elucidate the impact of ER-mitochondria signaling on PD pathogenesis and its associated impairments.

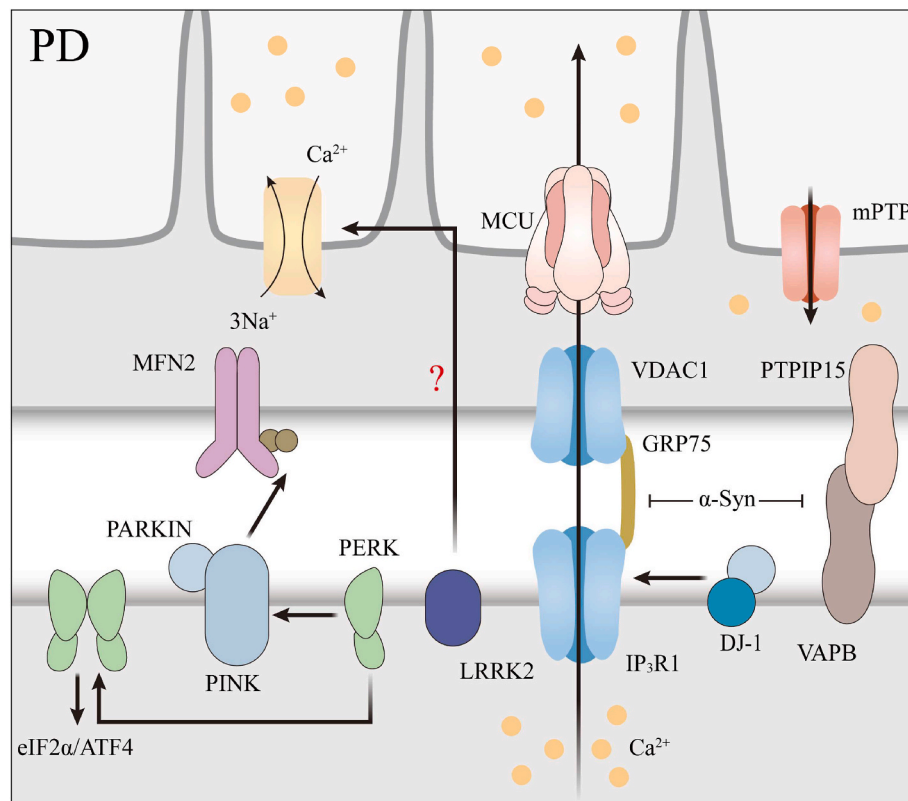


Fig. 3. Calcium homeostasis regulation in PD. Mutated forms of α -Syn lead to disruption of the VAPB/PTPIP51 interaction and impaired mitochondrial Ca^{2+} uptake. Furthermore, α -Syn can disrupt the interaction between IP₃R and GRP75, thereby hindering Ca^{2+} transfer. Disruption of communication between dysfunctional mitochondria and the ER can ameliorate signaling via the PERK branch of ER stress. Furthermore, PERK facilitates the ubiquitination and subsequent degradation of the mitochondrial fission factor MFN2 via PTEN-induced putative kinase 1 (PINK1) and PARKIN. Although there is a lack of evidence demonstrating the ability of leucine-rich repeat kinase 2 (LRRK2) to induce mitochondrial Ca^{2+} efflux through NCLX, the deficiency-induced inhibition of mitochondrial Ca^{2+} efflux by LRRK2 can be rescued through the upregulation of NCLX. DJ-1 can interact with the IP₃R-GRP75-VDAC complex. A deficiency of DJ-1 results in reduced ER-mitochondria contacts.

3.3. Amyotrophic lateral sclerosis

Amyotrophic lateral sclerosis is a progressive and fatal neurodegenerative disease characterized by the selective degeneration of upper and lower motor neurons (MNs). The progression of ALS is closely associated with impaired respiratory chain activity, mitochondrial dysfunction in MNs, and dysregulation of mitochondrial Ca^{2+} homeostasis.¹¹⁸ Mutations in several ALS-associated genes, including TAR DNA-binding protein 43 (TDP-43), superoxide dismutase 1 (SOD1), and C9orf72, have been reported to physically interact with the mitochondria and play pivotal roles in maintaining mitochondrial function.^{119–121}

Studies have reported that the rate of mitochondrial Ca^{2+} uptake doubled in mitochondria isolated from hindlimb muscles of 6-week-old SOD1 G93A mice.¹²² Aberrant mitochondrial Ca^{2+} uptake in skeletal muscle precedes other pathological events observed in ALS, potentially serving as a pivotal factor in the pathogenesis of ALS. The mechanism for accelerating Ca^{2+} absorption at this early stage has yet to be determined. One possibility may be regulating MCU complex-related protein expression.¹²³ In SOD1 G93A mice, MCU is overexpressed in embryonic spinal cord MNs and downregulated in adult MNs after ALS onset, possibly as a compensatory mechanism to reduce mitochondrial Ca^{2+} overload.¹²⁴ As ALS progresses, SOD1 G93A further accumulates within the mitochondria, eliciting mitochondrial membrane depolarization and subsequently diminishing mitochondrial Ca^{2+} uptake, decelerating mitochondrial dynamics and fostering mitochondrial ROS production¹²⁵ (Fig. 4). The spontaneous activity of mitochondrial mPTP in astrocyte processes is elevated in SOD1 G93A mice, which can be attributed to an imbalance in mitochondrial Ca^{2+} homeostasis and oxidative stress.³²

Misfolding, mislocalization, and hyperphosphorylation of TDP-43 are typical pathological features observed in ALS. TDP-43 and fused in

sarcoma (FUS) induce dissociation between VAPB and PTPIP51, thereby modulating Ca^{2+} transfer from the ER to the mitochondria via GSK3 β activation^{126,127} (Fig. 4). However, further investigations are warranted to elucidate the involvement of GSK3 β in MERCS. The VAPB P56S mutation has been reported to enhance its affinity for PTPIP51, thereby modulating Ca^{2+} shuttling between the ER and the mitochondria.¹²⁸ Ablation of either MFN2 or S1R leads to MERCS dysfunction, eliciting protein folding defects and inducing ER stress and UPR.¹²⁹ Notably, the C9orf72-ALS/FTD mouse model exhibits a significant early impairment in mitochondrial Ca^{2+} uptake ability, and direct binding of toxic PolyGR to ATP5A1 leads to mitochondrial functional defects.¹³⁰ PolyGA may actively contribute to mitochondrial dysfunction by sequestering C9orf72-GRP75 in ALS/FTD pathogenesis¹³¹ (Fig. 4). Despite the low abundance of PolyGA and predominant presence of TDP-43 in ALS pathology, exogenous overexpression of GRP75 also represents a potential therapeutic target for ALS, aiming to restore MERCS function and consequently reduce PolyGA aggregation.

In conclusion, MNs are susceptible to Ca^{2+} -induced mitochondrial dysfunction in ALS. The impact of ALS pathological protein mutations on mitochondrial Ca^{2+} homeostasis will result in a cascade of consequences, exposing cells to an energy crisis, oxidative damage, and ER stress, ultimately leading to MN death and weakened skeletal muscle contractile function.

3.4. Huntington's disease

Huntington's disease is an autosomal dominant neurodegenerative disorder resulting from the expansion of cellular CAG repeats within the coding sequence of the huntingtin gene (HTT). Mutant huntingtin proteins (mHTT) exhibit a propensity for proteolysis, misfolding, and

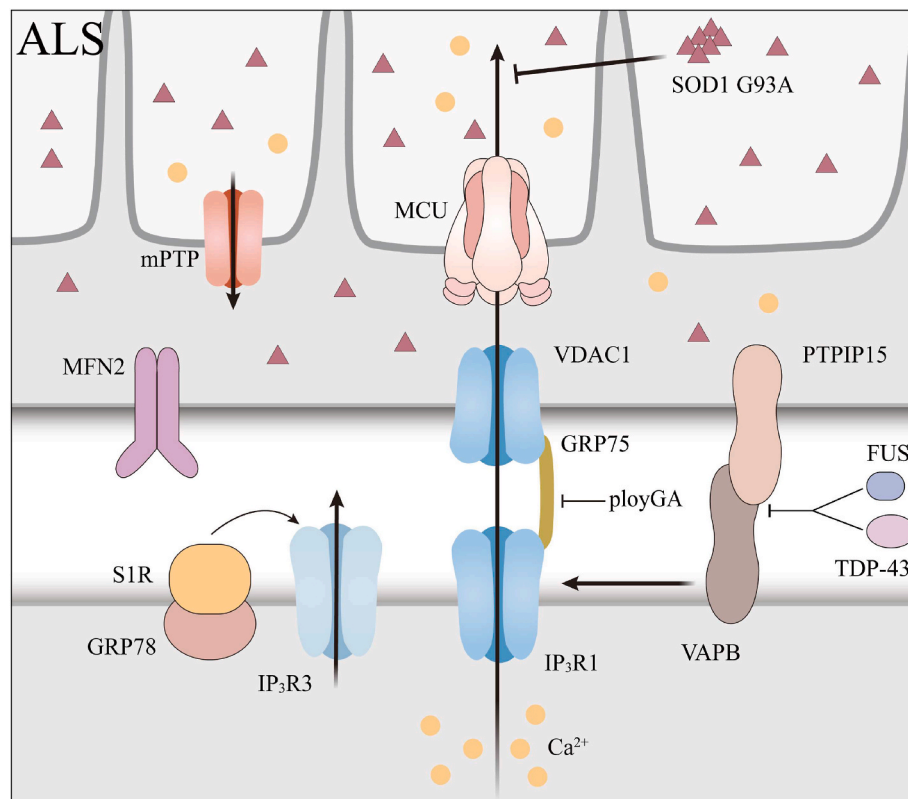


Fig. 4. Calcium homeostasis regulation in ALS. Superoxide dismutase 1 (SOD1) G93A further accumulates within mitochondria, eliciting mitochondrial membrane depolarization and subsequently diminishing mitochondrial Ca^{2+} uptake, decelerating mitochondrial dynamics. TAR DNA-binding protein 43 (TDP-43) and fused in sarcoma (FUS) induce dissociation between VAPB and PTPIP51, thereby modulating Ca^{2+} transfer from the ER to mitochondria. Ablation of either MFN2 or S1R leads to MAM dysfunction, eliciting protein folding defects and inducing ER stress and UPR. PolyGA may actively contribute to mitochondrial dysfunction by sequestering C9orf72-GRP75 in ALS/FTD pathogenesis.

aggregation. Clinically, HD is characterized by progressive impairments in motor, cognitive, and behavioral functions due to the degeneration of GABAergic neurons in the striatum. Similarly, examination of induced pluripotent stem cells and differentiated neural stem cells derived from HD patients have revealed aberrant mitochondrial morphology and impaired respiration, diminished activity of enzymes related to the TCA cycle, and initiation of apoptosis.¹³²

The primary mechanism underlying dysregulated Ca^{2+} homeostasis in HD is a mitochondrial dysfunction affecting both Ca^{2+} uptake and efflux. Previous studies have demonstrated that soluble N-terminal mHTT interacts with the OMM and IMM and directly interacts via TIM23 to disrupt mitochondrial protein import¹³³ (Fig. 5). The capacity for Ca^{2+} uptake increases with age in cultured striatal neurons of the yeast artificial chromosome (YAC)128 mouse model of HD, accompanied by an elevation in mitochondria-associated mHTT levels.¹³⁴ Over time, the mitochondria become overwhelmed with an excessive influx of Ca^{2+} , leading to the opening of mPTP and subsequent release of cytochrome *c* into the cytoplasm, thereby triggering activation of the caspase-mediated intrinsic apoptotic pathway.¹³⁵ Previous studies conducted by Tang et al. have demonstrated the in vitro and in vivo interaction between mHTT and IP₃R1 via the HTT-associated protein 1A (HAP1A) (Fig. 5). Furthermore, the authors provided evidence that the N-terminus of mHTT contains 138 glutamine repeats, which interacts with IP₃R1 and HAP1A, enhances the sensitivity of IP₃R1 to IP₃.¹³⁶ The RyR channels are modulated by post-translational modifications, including oxidation by ROS and phosphorylation by PKA. The increased permeability of RyR has been found in the brain, heart, and diaphragm of both HD patients and a murine model of HD (Q175).¹³⁷ RyR leakage leads to cognitive impairment, decreased parasympathetic tone associated with cardiac arrhythmias, and reduced diaphragmatic contractility in a murine model of HD (Q175).¹³⁷ However, the precise underlying mechanism of action remains to be fully elucidated.

The rapid absorption of Ca^{2+} into the mitochondrial matrix is facilitated by the enhanced activity of ER Ca^{2+} release channels (IP₃R1, RyR), supporting the regulation of HD-associated mitochondrial Ca^{2+}

homeostasis through ER-mitochondria contacts. Research has demonstrated that mHTT decreases the expression of S1R in neuronal PC6-3 cells, and the selective S1R agonist PRE084 enhances the expression of S1R, SOD1, and SOD2 in cellular models, suggesting that improved S1R function has a protective effect in HD¹³⁸ (Fig. 5). Although the molecular mechanism by which MERCS proteins regulate mitochondrial Ca^{2+} homeostasis remains elusive, activating S1R is essential for maintaining ER-mitochondria contacts and ensuring mitochondrial Ca^{2+} homeostasis in HD. A significant reduction in MERCS-related proteins IP₃R3 and GRP75 protein levels was observed in the striata of two HD mouse models and the human HD.¹³⁹ However, a decrease in MFN2 was only observed in the human HD putamen. In summary, Ca^{2+} homeostasis imbalance in HD depends on the intrinsic mitochondrial Ca^{2+} buffering capacity and the integrity of the MERCS.

3.5. Spinocerebellar ataxia

Spinocerebellar ataxias represent over 40 autosomal dominant and neurodegenerative disorders characterized by impaired balance and motor coordination arising from cerebellar dysfunction and its associated efferent connections. Disruption of the Ca^{2+} signaling pathway in cerebellar Purkinje cells is intricately related to the pathogenesis of SCA. Abnormal Ca^{2+} homeostasis in Purkinje cells activates molecular cascades that lead to cell death, cerebellar atrophy, and ataxia, characteristic of many SCAs.¹⁴⁰

The most prevalent form of SCA arises from an aberrant expansion of CAG repeats in members of the ataxin (ATXN) gene family, leading to the elongation of the polyglutamine (polyQ) region within the associated protein. By contrast, other forms of SCA arise from single missense mutations or deletions in genes encoding proteins typically associated with Ca^{2+} signaling cascades in Purkinje cells. For example, point mutations in IP₃R1, a protein related to the MERCS, contribute to SCA15/16 and SCA29, highlighting their crucial role in regulating mitochondrial Ca^{2+} homeostasis across various neurodegenerative disorders.^{140,141} SCA2 and SCA3 are two prevalent polyQ diseases caused by CAG repeat expansions in the genes encoding the ATXN2 and ATXN3 proteins.

In experiments conducted on the planar lipid bilayer model system, utilizing single-channel recordings of an IP₃R1 co-expressed with a polyQ-expanded ATXN2, it has been demonstrated that the ATXN2^{mut} protein specifically binds to the COOH terminus of the IP₃R1.¹⁴² In the SCA2-58Q Purkinje cells, IP₃R1 significantly augmented ER Ca^{2+} release and induced mitochondrial Ca^{2+} overload, which was attenuated by the RyR inhibitor dantrolene¹⁴² (Fig. 6). An excessive release of Ca^{2+} from the IP₃R1-sensitive Ca^{2+} pool may play a pivotal role in the impairment of Purkinje cell function in SCA2.

SCA3, also known as Machado-Joseph disease (MJD), is caused by an unstable expansion of the CAG repeat sequence in the ATXN3 gene. Similar to SCA2, the ATXN3^{mut} in SCA3 exhibits physical interaction with IP₃R1 expressed in neurons, thereby augmenting Ca^{2+} release and potentially triggering apoptosis of Purkinje cells.¹⁴³ Interestingly, in SCA3-79Q transgenic mice, both microarray analysis and real-time RT-PCR assays revealed a significant downregulation of the gene encoding IP₃R1.¹⁴⁴ These contradictory findings suggest that the pathogenesis of SCA3 is not a decrease or increase in IP₃R1 activity but instead reflects the loss of a precise regulation of its activity.

SCA15/16 and SCA29, caused by mutations in IP₃R1 encoded by ITPR1, share the same genetic target (Fig. 6). Current evidence suggests that haploinsufficiency of ITPR1 may underlie the pathogenesis of SCA15/16, while heterozygous missense mutations in ITPR1 disrupt the activity of IP₃R1 channels, leading to SCA29.^{145,146} Carbonic anhydrase-related protein VIII (CARP) is an IP₃R1-regulated protein encoded by CA8 and is abundantly expressed in cerebellar Purkinje cells. Notably, the SCA29 mutation V1538M within the CARP-binding site of IP₃R1 completely abolished its interaction with CARP and the inhibitory effect of CARP on IP₃R1.¹⁴⁶ Collectively, while SCA15/16 and SCA29

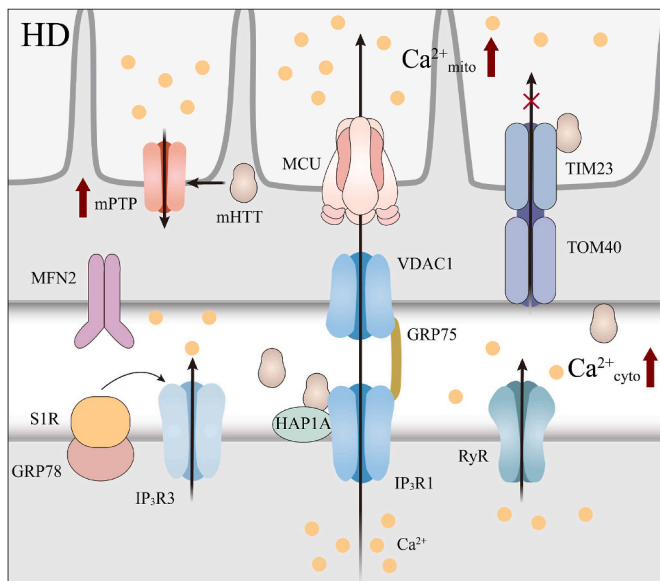


Fig. 5. Calcium homeostasis regulation in HD. Soluble N-terminal mutant huntingtin proteins (mHTT) interacts with the OMM and IMM and directly interacts through TIM23 to disrupt mitochondrial protein import. In vitro and in vivo interaction between mHTT and IP₃R1 through HTT-associated protein 1A (HAP1A). In HD model (Q175), RyR channel oxidation leads to ER Ca^{2+} leakage. Improved S1R function has a protective effect in HD. Over time, mitochondria become overwhelmed with an excessive influx of Ca^{2+} , leading to the opening of mPTP, thereby triggering activation of the apoptotic pathway.

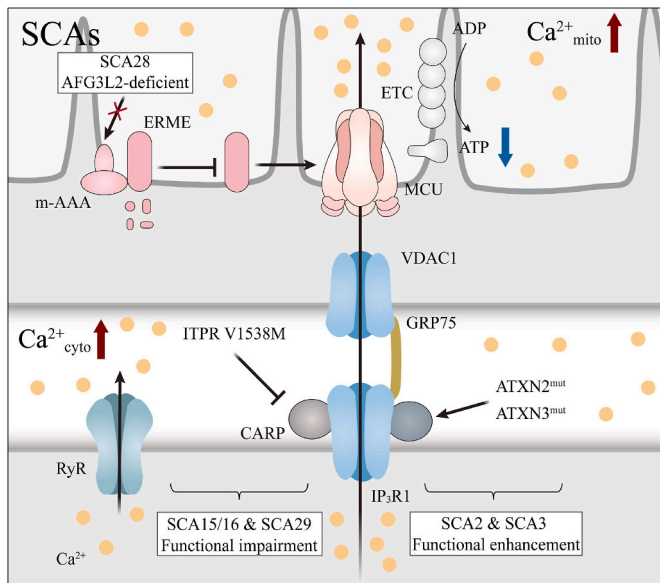


Fig. 6. Calcium homeostasis regulation in SCAs. ATXN2mut and ATXN3mut in SCA2 and SCA3 exhibit physical interactions with IP3R1, thereby increasing ER Ca^{2+} release and inducing mitochondrial Ca^{2+} overload. The SCA29 mutation V1538M within the Carbonic anhydrase-related protein VIII (CARP)-binding site of IP3R1 completely abolished its interaction with CARP and the inhibitory effect of CARP on IP3R1. Impaired degradation of EMRE in AFG3 ATPase family gene 3-like 2 (AFG3L2)-deficient mitochondria results in the dysregulation of MCU complexes and enhances neuronal susceptibility to Ca^{2+} -induced cell death.

arise from IP₃R1 loss of function due to either dominant negative impact (SCA29) or haploinsufficiency (SCA15/16), the converse is observed in SCA2 and SCA3, indicating the loss of precise regulation of IP₃R1 activity during the pathogenesis of SCA, ultimately leading to Purkinje cell dysfunction and death (Fig. 6).

Finally, mutations in the AFG3 ATPase family gene 3-like 2 (AFG3L2), which encodes a protease subunit with an exposed catalytic structure to matrix AAA (m-AAA), are associated with SCA28 and are linked to the loss of Purkinje cells¹⁴⁷ (Fig. 6). Missense mutations in AFG3L2 and deletion of AFG3L2 lead to aberrant mitochondrial morphology, impaired respiratory chain complexes, and diminished mitochondrial ATP production.¹⁴⁸ Mechanistically, the degradation of unassembled EMRE by m-AAA protease ensures the binding of EMRE-MICU assembly intermediates to MCU subunits to maintain current mitochondrial Ca^{2+} homeostasis.¹⁴⁹ Impaired degradation of EMRE in AFG3L2-deficient mitochondria results in the dysregulation of MCU complexes and enhances neuronal susceptibility to Ca^{2+} -induced cell death.¹⁵⁰ Strikingly, deletion of MCU in Purkinje cells lacking AFG3L2 does not suppress neurodegeneration or neuroinflammatory responses, indicating that the disturbance of mitochondrial Ca^{2+} homeostasis is not the sole cause of neuronal death in SCA28.¹⁵⁰

In conclusion, the dysregulation of cerebellar Purkinje cell Ca^{2+} homeostasis plays a pivotal role in the pathogenesis of motor dysfunction in SCA. However, further elucidation of the impact of different SCA mutations on mitochondrial Ca^{2+} buffering is required. Consequently, targeting specific proteins that maintain mitochondrial Ca^{2+} homeostasis could be a promising therapeutic strategy to prevent neuronal death caused by Purkinje cell Ca^{2+} accumulation of various SCAs.

4. Therapies targeting mitochondrial Ca^{2+}

Given that dysregulation of mitochondrial Ca^{2+} homeostasis can trigger neurodegeneration through apoptosis/necrosis, it is crucial to identify targets that maintain Ca^{2+} balance and preserve mitochondrial

function to develop disease-modifying drugs in neurodegenerative disorders. Dysregulation of Ca^{2+} homeostasis in mitochondria may arise from increased uptake mediated by the MCU complex, decreased efflux facilitated by NCLX, altered buffering capacity for Ca^{2+} , or enhanced ER-mitochondrial Ca^{2+} transfer. Table 1 summarizes various drugs and compounds that modulate different aspects of mitochondrial Ca^{2+} homeostasis in neurodegenerative diseases.^{90,151–171}

In addition to the treatments mentioned above that modulate mitochondrial Ca^{2+} homeostasis, there are ongoing investigations into several drugs or compounds with potential therapeutic value. The compound MCU-i11, discovered through high-throughput screening, influences MCU activity through MICU1.¹⁷² However, MCU-i11's efficacy is currently restricted to cultured cells, with no significant in vivo efficacy reported. Before progressing to clinical trials, modifications to MCU-i11 are necessary to enhance its bioavailability and mitigate potential side effects. The VDAC1-specific inhibitor VBIT-4 effectively restores cognitive function in AD-like mouse models by preventing mitochondrial metabolic dysfunction, apoptosis, and neuroinflammation.¹⁷³ However, further experimental evidence is needed to support the use of apoptosis inhibitors for treating AD. Beyond its role in transmitting pro-apoptotic Ca^{2+} signals to mitochondria, VDAC1 is also involved in accumulating mutant SOD1.¹⁷⁴ The ability of VBIT-12 to rescue cell death induced by mutant SOD1 in neuronal cultures suggests that it could be a potential target for therapy. Furthermore, Andy Mark's team has developed a novel compound called Rycal, which stabilizes RyR1-S2844D and promotes interactions with calstain, preventing ER Ca^{2+} leakage and reducing the progression of Ca^{2+} in tremor pathophysiology.¹⁷⁴

Notably, current therapies are increasingly targeting specific pathological proteins in neurodegenerative diseases, yet there remains a limited number of therapeutics focused on regulating mitochondrial Ca^{2+} homeostasis.^{164,165,168,175} Future therapeutic potential for neurodegenerative diseases will be significantly enhanced by advancements in mitochondrial transplantation, mesenchymal stem cell therapy, gene therapy, and drug research targeting mitochondrial Ca^{2+} homeostasis and MERCS-related genes.¹⁷⁶

5. Conclusion and future perspective

In this review, we emphasize the pivotal role of $\text{Ca}^{2+}_{\text{mit}}$ in both physiological and pathophysiological contexts. Mitochondria tightly regulate neuronal lifespan through precise control of ATP production and Ca^{2+} homeostasis. Impaired mitochondrial function, manifested by increased ROS generation, altered mitochondrial membrane potential, decreased mitochondrial ATP synthesis and mPTP opening, leads to severe neurodegeneration and even neuronal death. This review also provides a comprehensive overview of the intricate relationship between dysregulation of mitochondrial Ca^{2+} homeostasis and the underlying pathophysiological mechanisms implicated in neurodegeneration. Targeting $\text{Ca}^{2+}_{\text{mit}}$ dynamics and its associated regulators holds great promise as a potential preventive strategy for various neurodegenerative disorders, including AD, PD, ALS, HD, and SCA. Furthermore, we present an overview of targeted therapeutic interventions aimed at modulating different $\text{Ca}^{2+}_{\text{mit}}$ regulatory proteins in the context of neurodegenerative diseases (Table 1).

Collectively, the key regulatory factors implicated in mitochondrial Ca^{2+} influx and efflux, particularly those involving MERCS, represent potential therapeutic targets. Pharmacological agents or compounds that selectively inhibit or modulate these crucial proteins offer precise control over mitochondrial Ca^{2+} homeostasis. Future research should focus on investigating intact neuronal circuits and neuron-glia communication using experimental models that faithfully replicate human neurodegenerative disease pathology. This approach will elucidate how mitochondrial Ca^{2+} homeostasis and metabolic processes influence disease progression, providing invaluable insights.⁸⁸ Furthermore, the development of drugs that selectively activate or

Table 1
Therapies targeting mitochondrial calcium in neurodegenerative diseases.

Site	Treatment	Disease	Model	Effects	References
MCU	Ru360	AD	Applying naturally secreted soluble A β to the mouse brain surface	Decreased mitochondrial Ca ²⁺ uptake	89
		AD	Primary mouse microglia exposed to A β	Attenuated apoptosis, mitochondrial ROS production, and ER stress	151
		PD	PD neurotoxin induces preformed fibrils of α -Syn in mice	Reduced mitochondrial calcium uptake, oxidative stress, and cytochrome <i>c</i> release	152
$\Delta\Psi_m$	FCCP	ALS	SOD1 G93A transgenic mice	Reduced ROS generation and cell apoptosis	153
		AD	Cultured rat hippocampal neurons exposed to A β	Decreased mitochondrial Ca ²⁺ uptake and neuronal death via mitochondrial depolarization	154
	2,4-Dinitrophenol (DNP)	HD	N171-82Q HD mice	A low dose of DNP reduces oxidative stress	161
		PD	An acute mouse model of PD	mPTP-induced motor dysfunction was alleviated in the DNP-treated mice	162
mPTP	Berberine	AD	Mouse hippocampal neurons exposed to A β	Maintenance of $\Delta\Psi_m$, prevention of ATP decrease	154
		AD	Fibroblasts isolated from sAD patients	Decreased mitochondrial superoxide levels, improved mitochondrial Ca ²⁺ dysregulation	156
	CsA	ALS	SOD1 G93A transgenic mice	Blocked the opening of mitochondrial mPTP, reduced ROS production and cell death	153
		PD	Rotenone-induced SH-SY5Y cell PD model	Reduced loss of dopaminergic neurons in the substantia nigra of the striatum	157
NCLX	CGP37157	AD	Primary rat cortical neurons exposed to acetylcholine, KCl, and NMDA	Decreased mitochondrial Ca ²⁺ overload	[158,159]
MAMs	Methyl vitamin B12	AD	PC12 cells exposed to A β ₂₅₋₃₅	Reduced ER-mitochondria Ca ²⁺ flux through IP ₃ R, apoptosis, and cell death	160
RyR	Dantrolene	AD	APP(swe)-expressing (Tg2576) mice	RyR-mediated decrease in ER Ca ²⁺ release and reduced A β load in neurons	163
		HD	Striatal YAC128 HD mouse model	Reduced NeuN-positive striatal neuron death and mHTT accumulation in the nucleus	164
		SCA2	Generation of SCA2-58Q transgenic mice	Mitigates age-dependent motor deficits and Purkinje cell loss	142
S1R	PRE084	SCA3	Generation of SCA3-YAC-84Q transgenic mice	Improved motor coordination and neuronal loss	143
		AD	A β ₂₅₋₃₅ -induced mouse model of AD	Attenuated A β -induced ER stress-related neurotoxicity and promoted neurogenesis	165
		PD	MPTP-induced PD mouse model	Promoted PINK1/PARKIN-mediated mitophagy and reduced neuron loss	166
		HD	Neuronal PC6.3 cells expressing N-terminal mHTT	Reduced ROS levels and activated NF- κ B signaling to counteract the mHTT in neurons	138
	Pridopidine	ALS	TDP43 G348C zebrafish ALS model	Stimulated mitochondrial respiration and ATP production	167
		ALS	SOD1 G93A transgenic mice	Reduced MN death and SOD1 aggregation and restored axonal transport of mitochondria	168
Mitochondrial Ca ²⁺ transient	Riluzole	HD	Striatal YAC128 HD mouse model	Improved ER-mitochondria interactions and increased ATP production	169
		ALS	SH-SY5Y G93A neuroblastoma cell lines	Reduced MN death and oxidative stress by inhibiting mitochondrial Ca ²⁺ signaling	170
		SCA2/3	SCA2- and SCA3-iPSC-derived neurons	Reduced cell death and improved mitochondrial function	171

inhibit components of the MCU complex, NCLX, and MERCS presents challenges in terms of ensuring clinical safety and achieving drug delivery specificity to avoid unintended effects on mitochondrial Ca²⁺ homeostasis in unaffected tissues. Additionally, dynamic regulation of mitochondrial Ca²⁺ homeostasis during the progression of various neurodegenerative disorders is essential for early diagnosis and effective treatment.

CRedit authorship contribution statement

Jianxu Sun: Writing – original draft. **Ge Gao:** Writing – review & editing. **Sitong Wang:** Writing – review & editing. **Hongmei Liu:** Writing – review & editing. **Tie-Shan Tang:** Writing – review & editing, Funding acquisition.

Data availability

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Declaration of competing interest

The authors declare no competing interests.

Abbreviations

Abbreviation	Meaning
A β	Amyloid beta
AD	Alzheimer's disease
AFG3L2	AFG3 ATPase family gene 3-like 2
α -KGDH	α -ketoglutarate dehydrogenase
α -Syn	α -Synuclein
ATF4	Activating transcription factor 4
ATXN	Ataxin
ALS	Amyotrophic lateral sclerosis
APOE	Apolipoprotein E
APP	Amyloid beta precursor protein
BACE1	Beta-site APP cleaving enzyme 1
CARP	Carbonic anhydrase-related protein
Ca ²⁺ +mito	Mitochondrial Ca ²⁺
Ca ²⁺ +cyto	Cytoplasm Ca ²⁺
CNX	Calnexin
eIF2 α	α subunit of eukaryotic initiation factor 2
EIF2AK3/PERK	Eukaryotic translation initiation factor 2 α kinase 3
ER	Endoplasmic reticulum
ERME	Essential MCU regulator
ERN1/IRE1	ER to nucleus signaling 1
ETC	Electron transport chain
FUS	Fused in sarcoma
GSK3 β	Glycogen synthase kinase-3 β
HAP1A	HTT-associated protein 1A
HD	Huntington's disease
HSPA9/GRP75	Heat shock protein family A member 9
LETM1	Leucine zipper EF-hand-containing transmembrane
LRRK2	Leucine-rich repeat kinase 2
ICDH	Isocitrate dehydrogenase
IMM	Inner mitochondrial membrane
IMS	Intermembrane space
IP3Rs	Inositol 1,4,5-trisphosphate receptors 1 to 3
$\Delta\Psi_m$	Mitochondrial membrane potential
MARCHF5/MITOL	Membrane-associated ring-CH-type finger 5
MERCS	Mitochondria-endoplasmic reticulum contact sites
m-AAA	Matrix AAA
MCU	Mitochondrial Ca ²⁺ uniporter
MCUR1	MCU regulator 1
MFN	Mitofusin
mHTT	Mutant huntingtin proteins
MICU	Mitochondrial Ca ²⁺ uptake
MJD	Machado-Joseph disease
mPTP	Mitochondrial permeability transition pore
NCLX	Na ⁺ /Ca ²⁺ exchanger
NFTs	Neurofibrillary tangles
OMM	Outer mitochondrial membrane
OXPHOS	Oxidative phosphorylation system
PACS2	Phosphofuranic acid cluster sorting protein 2
PD	Parkinson's disease
PDH	Pyruvate dehydrogenase
PDZD8	PDZ Domain Containing 8
PKA	Protein kinase A
PKC	Protein kinase C
polyQ	Polyglutamine
PRKN/PARKIN	Parkin RBR E3 ubiquitin-protein ligase
PS1/PSEN1	Presenilin
PTEN1	PTEN-induced putative kinase 1
RMDN3/PTPIP51	Regulatory microtubule dynamics protein 3
ROS	Reactive oxygen species
RyRs	Ryanodine receptors
S1R	Sigma-1 receptor
S1T	ATPase sarcoplasmic reticulum/ER Ca ²⁺ transporter 1
SC	Supercomplexes
SCA	Spinocerebellar Ataxia
SCAF1	SR-Related CTD assembly factor 1
SEPN1	Selenoprotein N
SERCAs	Sarco-ER Ca ²⁺ transport ATPases
SNpc	Substantia nigra pars compacta
SOCE	Store-operated Ca ²⁺ entry
SOD1	Superoxide dismutase 1
TCA	Tricarboxylic acid
TDP-43	TAR DNA-binding protein 43
TIM23	Translocase of the inner mitochondrial membrane 23
TMCO1	Transmembrane and coiled-coil domains 1
TMX1	Thioredoxin-related transmembrane protein 1
TOM40	Translocase of the outer mitochondrial membrane 40
UPR	Unfolded protein response

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VAPB	Vesicle-associated membrane protein-associated protein B
VDACs	Voltage-dependent anion channels
YAC	Yeast artificial chromosome

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